ECE 340: Semiconductor Devices

Lecture 16: Diffusion

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Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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(Recap) Diffusion Current

$$\phi_n(x_o) = \frac{\bar{l}}{2\bar{t}}(n_1 - n_2)$$

Since $\bar{l}$ is small: $n(x) - n(x+\Delta x) = (n_1 - n_2)\frac{\Delta x}{\bar{l}}$ and thus $(n_1 - n_2) = \frac{n(x) - n(x+\Delta x)}{\Delta x}\bar{l}$

In the limit of small Δx: $\phi_n(x) = \frac{\bar{l}}{2\bar{t}} \lim_{\Delta x \to 0}\left(\frac{n(x) - n(x+\Delta x)}{\Delta x}\right) = \frac{-\bar{l^2}}{2\bar{t}}\frac{dn(x)}{dx}$

The term $\frac{\bar{l^2}}{2\bar{t}}$ is the electron diffusion coefficient D_n with units cm^2/s

$$D_n = \frac{\bar{l^2}}{2\bar{t}}$$

A similar relation can be derived for hole motion. So:

$$\phi_n(x) = -D_n\frac{dn(x)}{dx}$$

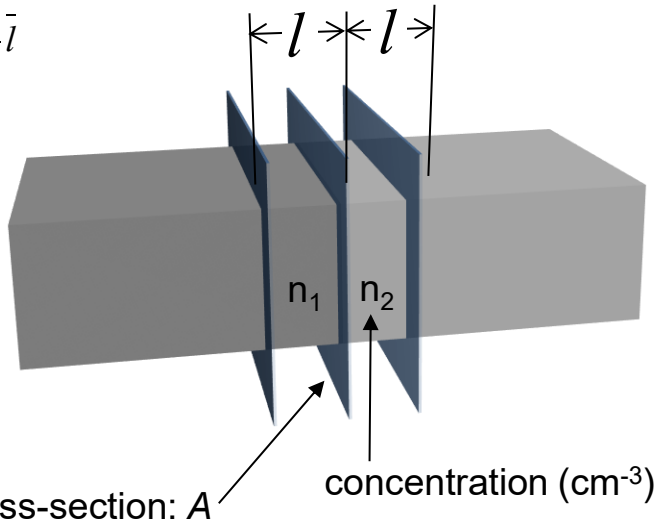
$$\phi_p(x) = -D_p\frac{dp(x)}{dx}$$

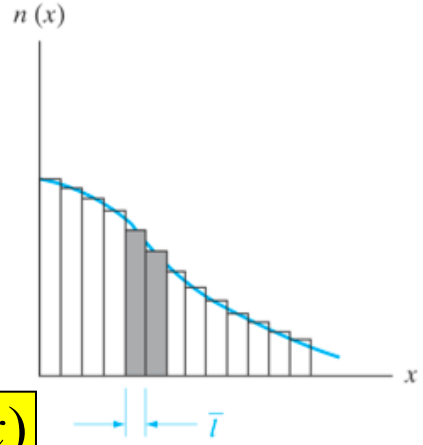
Electron and hole flux can be expressed as current:

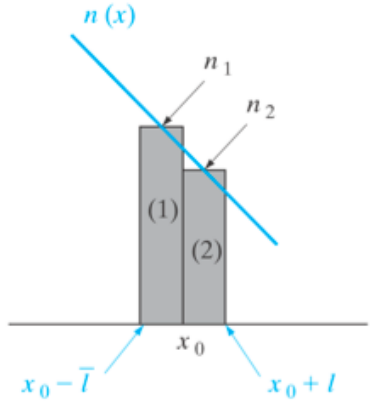
$$J_n(diff.) = -D_n\frac{dn(x)}{dx} \times (-q) = qD_n\frac{dn(x)}{dx}$$

$$J_p(diff.) = -D_p\frac{dp(x)}{dx} \times (+q) = -qD_p\frac{dp(x)}{dx}$$

$$J_n(x) = qD_n\frac{dn(x)}{dx} \text{ and } J_p(x) = -qD_p\frac{dp(x)}{dx}$$







(Recap) Total Current: Drift and Diffusion

$$J_n(x) = J_n(drift) + J_n(diff)$$

$$\rightarrow J_n(x) = qn(x)\mu_n\mathcal{E}(x) + qD_n\frac{dn(x)}{dx}$$

$$J_p(x) = J_p(drift) + J_p(diff)$$

$$\rightarrow J_p(x) = qp(x)\mu_p\mathcal{E}(x) - qD_p\frac{dp(x)}{dx}$$

$$\rightarrow J(x) = J_n(x) + J_p(x)$$

$$J(x) = qn(x)\mu_n\mathcal{E}(x) + qD_n\frac{dn(x)}{dx} + qp(x)\mu_p\mathcal{E}(x) - qD_p\frac{dp(x)}{dx}$$

Einstein Relationship

Band Diagram Illustration: Carriers Moving in an Electric Field

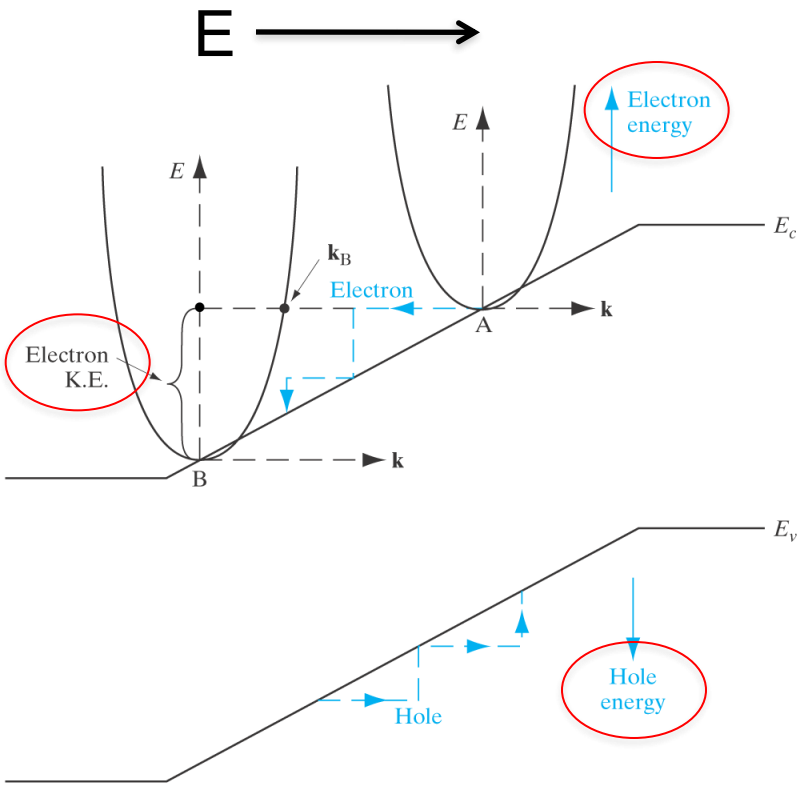


Figure 3.9

Superimposition of the (E,$\mathbf{k}$) band structure on the E-versus-position simplified band diagram for a semiconductor in an electric field. Electron energies increase going up, while hole energies increase going down. Similarly, electron and hole wave vectors point in opposite directions and these charge carriers move opposite to each other, as shown.

From Previous Lectures:

- The electron (hole) kinetic energy is zero at the conduction (valence) band edge
- Total energy = Kinetic Energy + Potential Energy
- The Potential Energy is given by the relative position of the band edge

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Band Diagrams With Electric Fields

- As the electron is accelerated in the field, it loses potential energy and gains kinetic energy
- When an electric field is present, the bands are tilted to reflect the loss in potential energy as the electron moves in the direction opposite to the field
- The electric field can be externally applied or internally generated by doping gradients

$\mathcal{V}(x) \equiv$ Electrostatic Potential

$E(x) \equiv$ Electron Potential Energy

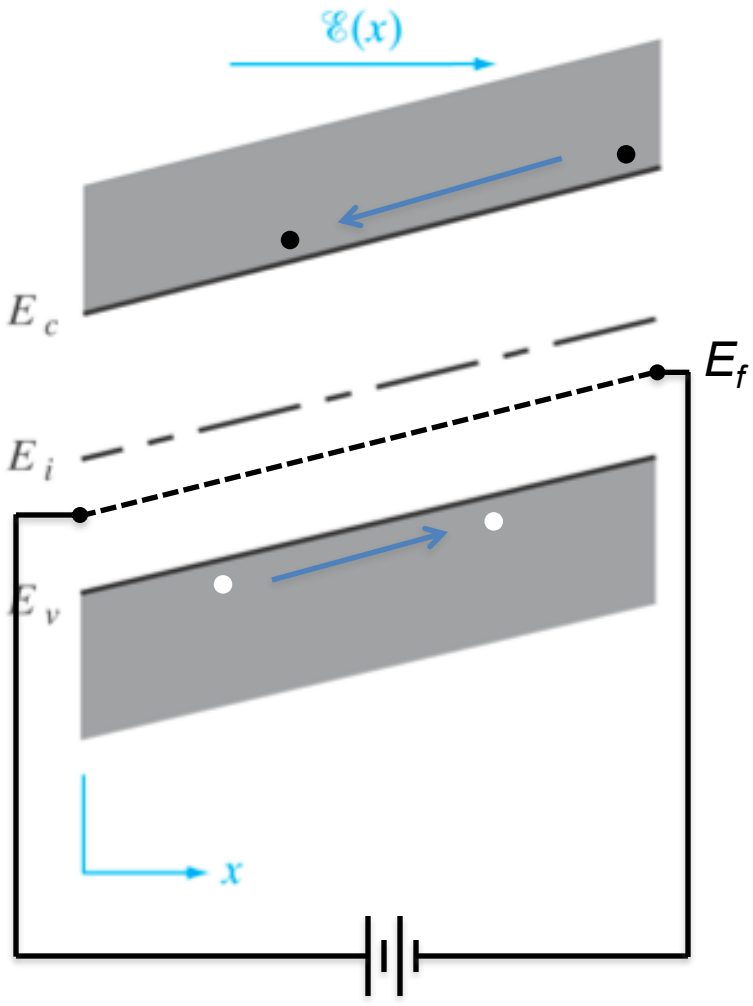
$\mathcal{E}(x) \equiv$ Electric Field

$$\mathcal{V}(x) = \frac{E(x)}{(-q)} \text{ and } \mathcal{E}(x) = -\frac{d\mathcal{V}(x)}{dx}$$

Choosing E_i as a spatial reference:

$$\mathcal{E}(x) = -\frac{d\mathcal{V}(x)}{dx} = -\frac{d}{dx}\left[\frac{E_i}{(-q)}\right] = \frac{1}{q}\frac{dE_i}{dx}$$

Externally Applied Field:



Derivation of Einstein Relationship

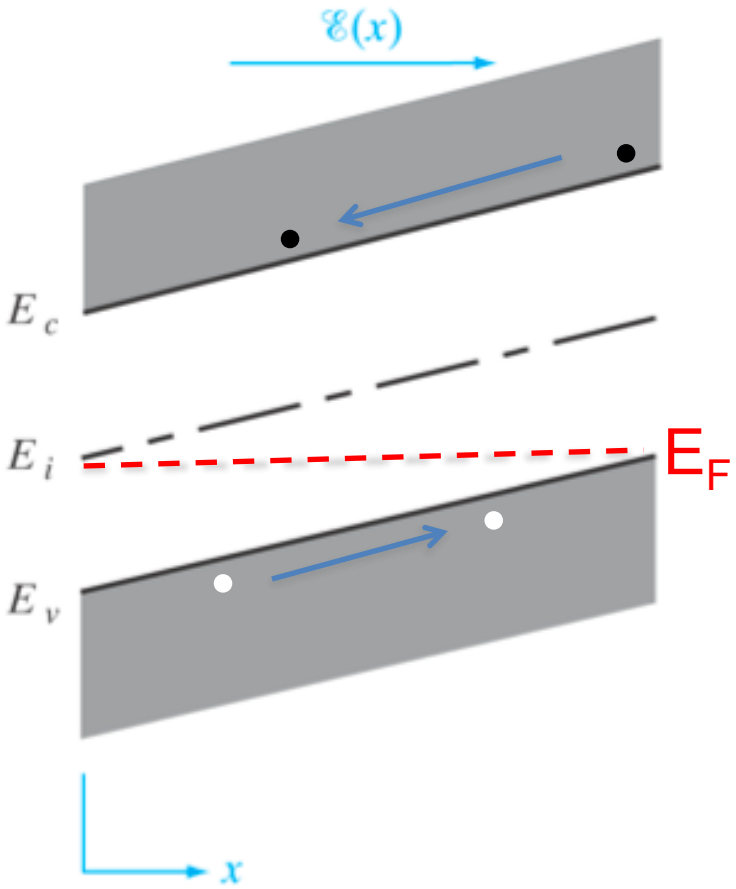
- E field can be generated internally.
- How can we generate the linear field?

$$E_i(x) = E_i(0) + q\mathcal{E}_0 x, \qquad \mathcal{E}_0 = \frac{1}{q}\frac{dE_i}{dx} > 0.$$

$$p_0(x) = n_i e^{|E_i(x) - E_F|/k_B T}$$

- Exponentially increasing doping concentration along *x* can generate linear field.
- Fermi level is invariant (equilibrium) → there is no net current (why?)

Internally Generated Field:



We can also have tilted bands with no externally applied potential!

Derivation of Einstein Relationship

$$J_p(x) = qp(x)\mu_p\mathcal{E}(x) - qD_p\frac{dp(x)}{dx} = 0$$

$$\text{so } \mathcal{E}(x) = \frac{D_p}{\mu_p}\frac{1}{p(x)}\frac{dp(x)}{dx}$$

$$p_0(x) = n_i e^{(E_i(x)-E_F(x))/kT}.$$

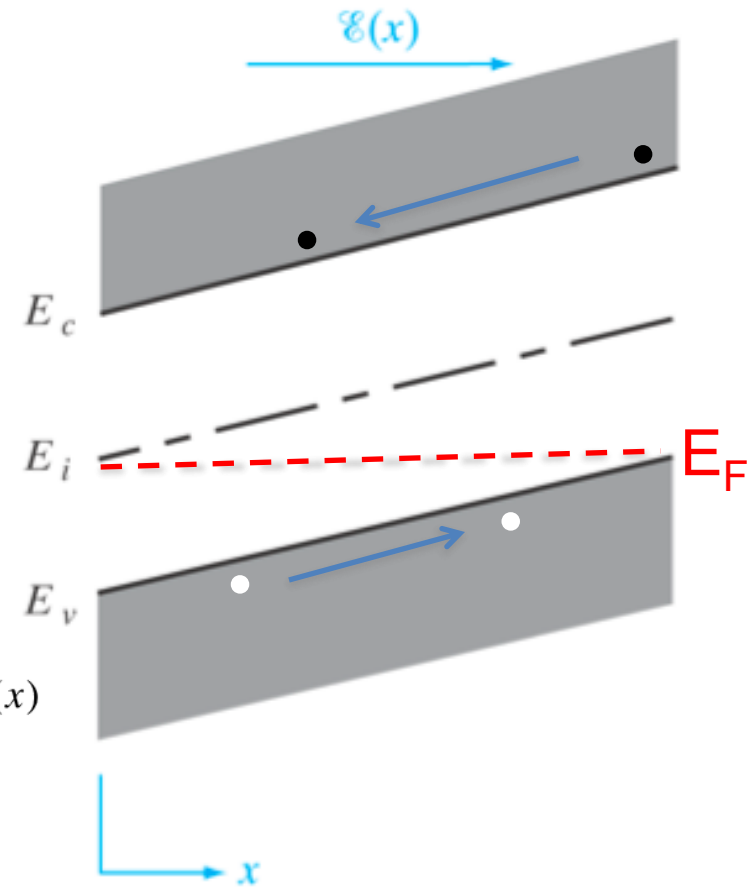
$$\text{using } \frac{d}{dx}e^{f(x)} = e^{f(x)}f'(x)$$

$$\frac{dp_0}{dx} = n_i e^{(E_i-E_F)/kT}\frac{d}{dx}\left[\frac{E_i - E_F}{kT}\right]$$

$$\mathcal{E}(x) = \frac{D_p}{\mu_p}\frac{1}{kT}\left(\frac{dE_i}{dx} - \frac{dE_F}{dx}\right) = \frac{D_p}{\mu_p}\frac{1}{kT}(q\mathcal{E}(x) - 0) \text{ since } \frac{1}{q}\frac{dE_i}{dx} = \mathcal{E}(x)$$

$$\text{so } 1 = \frac{D_p}{\mu_p}\frac{q}{kT} \text{ and } \boxed{\frac{D_p}{\mu_p} = \frac{kT}{q}}$$

Internally Generated Field:



We can also have tilted bands with no externally applied potential!

Key Points

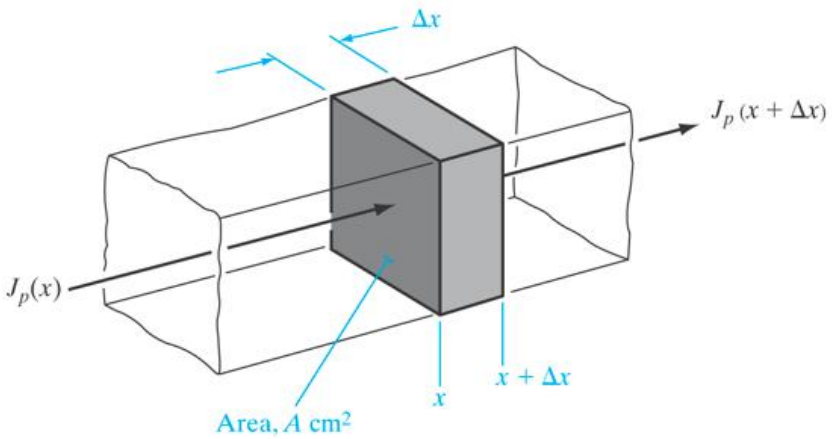
- The Einstein relation allows us to calculate either D or μ from a measurement of the other
- The Fermi level E_F is constant in equilibrium
- The intrinsic energy E_i is fixed relative to the band edges
- Material properties that cause E_i to move relative to E_F will create built-in fields that balance drift and diffusion
  - Doping concentration, doping type, bandgap (alloy composition), etc.

Diffusion and Recombination

The Continuity Equation

Description of Continuity Equation

Current Entering and Leaving a Volume Δx•A



Rate of Buildup = Increase in Concentration in Volume Δx•A per unit time - Recombination Rate

Holes:

$$\left.\frac{dp(x)}{dt}\right|_{x\to x+\Delta x} = \frac{1}{(+q)}\frac{J_p(x)-J_p(x+\Delta x)}{\Delta x} - \frac{\delta p}{\tau_p}$$

Electrons:

$$\left.\frac{dn(x)}{dt}\right|_{x\to x+\Delta x} = \frac{1}{(-q)}\frac{J_n(x)-J_n(x+\Delta x)}{\Delta x} - \frac{\delta n}{\tau_n}$$

- The current flowing out of a volume can be different from the current flowing into a volume.
- The **net increase in carrier concentration** per interval of time is **the rate entering, minus the rate leaving, minus the number that have recombined.** (assuming no generation)

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Electron and Hole Continuity Equations

Assuming No Generation:

Hole Continuity Equation

$$\left.\frac{dp(x)}{dt}\right|_{x\to x+\Delta x} = \frac{1}{(+q)}\frac{J_p(x)-J_p(x+\Delta x)}{\Delta x} - \frac{\delta p}{\tau_p}$$

as $\Delta x \to 0$

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial (p_0+\delta p)}{\partial t} = \boxed{\frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}}$$

Note, as $\Delta x \to 0$:

$$\frac{J_p(x+\Delta x)-J_p(x)}{\Delta x} = \frac{\partial J_p(x)}{\partial x}$$

Electron Continuity Equation

$$\left.\frac{dn(x)}{dt}\right|_{x\to x+\Delta x} = \frac{1}{(-q)}\frac{J_n(x)-J_n(x+\Delta x)}{\Delta x} - \frac{\delta n}{\tau_n}$$

as $\Delta x \to 0$

$$\frac{\partial n(x,t)}{\partial t} = \frac{\partial (n_0+\delta n)}{\partial t} = \boxed{\frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n}}$$

Electron and Hole Continuity Equations

With Generation considered:

Hole Continuity Equation

$$\left.\frac{dp(x)}{dt}\right|_{x\to x+\Delta x} = \frac{1}{(+q)}\frac{J_p(x)-J_p(x+\Delta x)}{\Delta x} - \frac{\delta p}{\tau_p} + G$$

as $\Delta x \to 0$

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p} + G$$

Electron Continuity Equation

$$\left.\frac{dn(x)}{dt}\right|_{x\to x+\Delta x} = \frac{1}{(-q)}\frac{J_n(x)-J_n(x+\Delta x)}{\Delta x} - \frac{\delta n}{\tau_n} + G$$

as $\Delta x \to 0$

$$\frac{\partial n(x,t)}{\partial t} = \frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n} + G$$

Continuity Equations

$$\frac{\partial n}{\partial t} = G_n - U_n + \frac{1}{q}\nabla \bullet \vec{J}_n$$

$$\frac{\partial p}{\partial t} = G_p - U_p - \frac{1}{q}\nabla \bullet \vec{J}_p$$

Divergence of a vector field

For a given volume of semiconductor, the rate change of carrier is the net effect of current flow into the volume and generation and recombination rates within the volume.

G_n : *electron generation rate*

G_p : *hole generation rate*

U_n : *electron recombination rate*

U_p : *hole recombination rate*